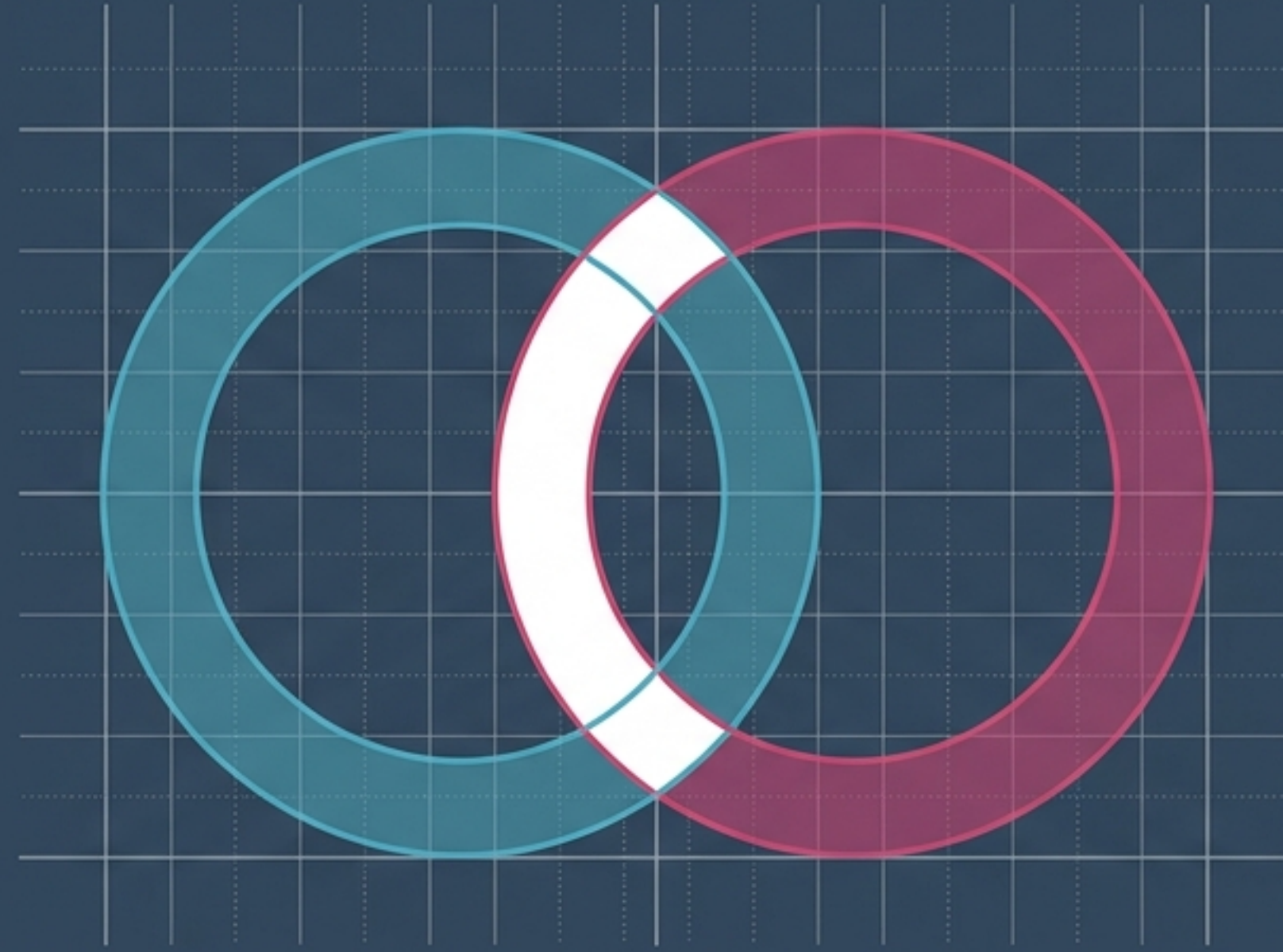


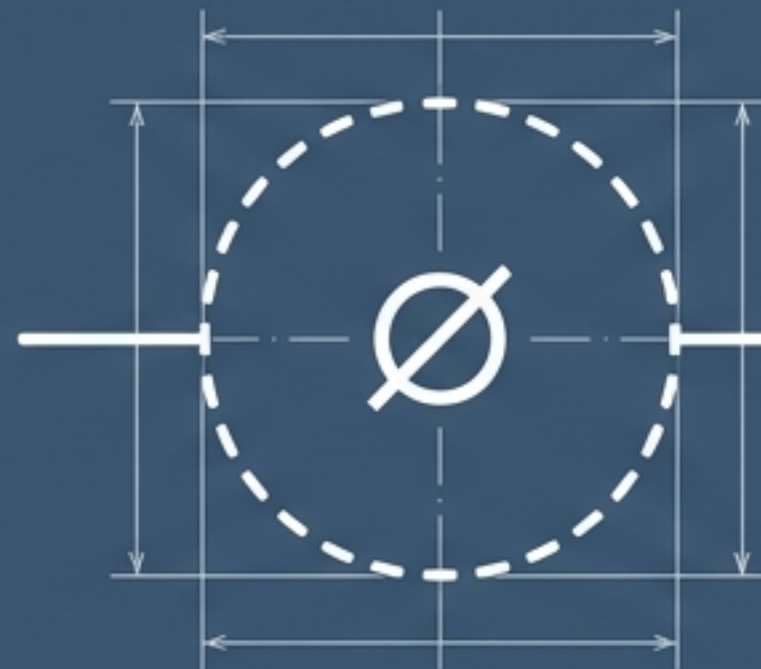


# THE ARCHITECTURE OF LOGIC

A Visual Guide to Sets, Operations, Partitions, and Relations

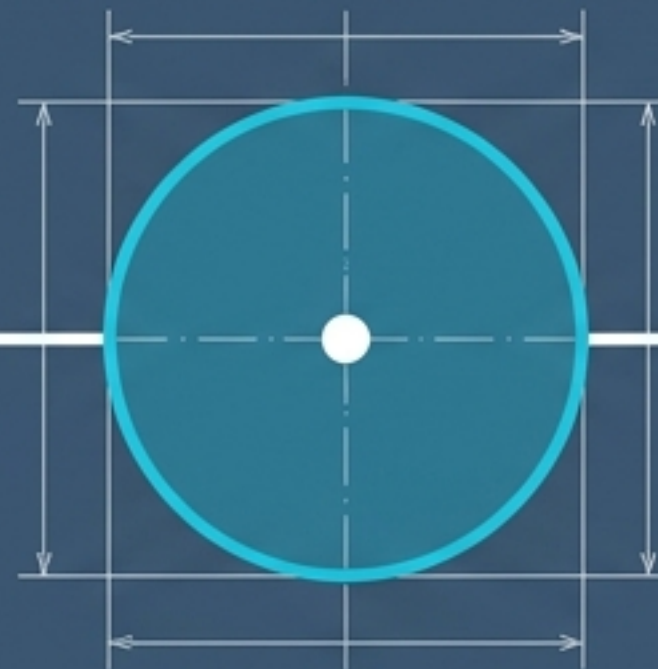
## Universal Set

## Empty (Null) Set

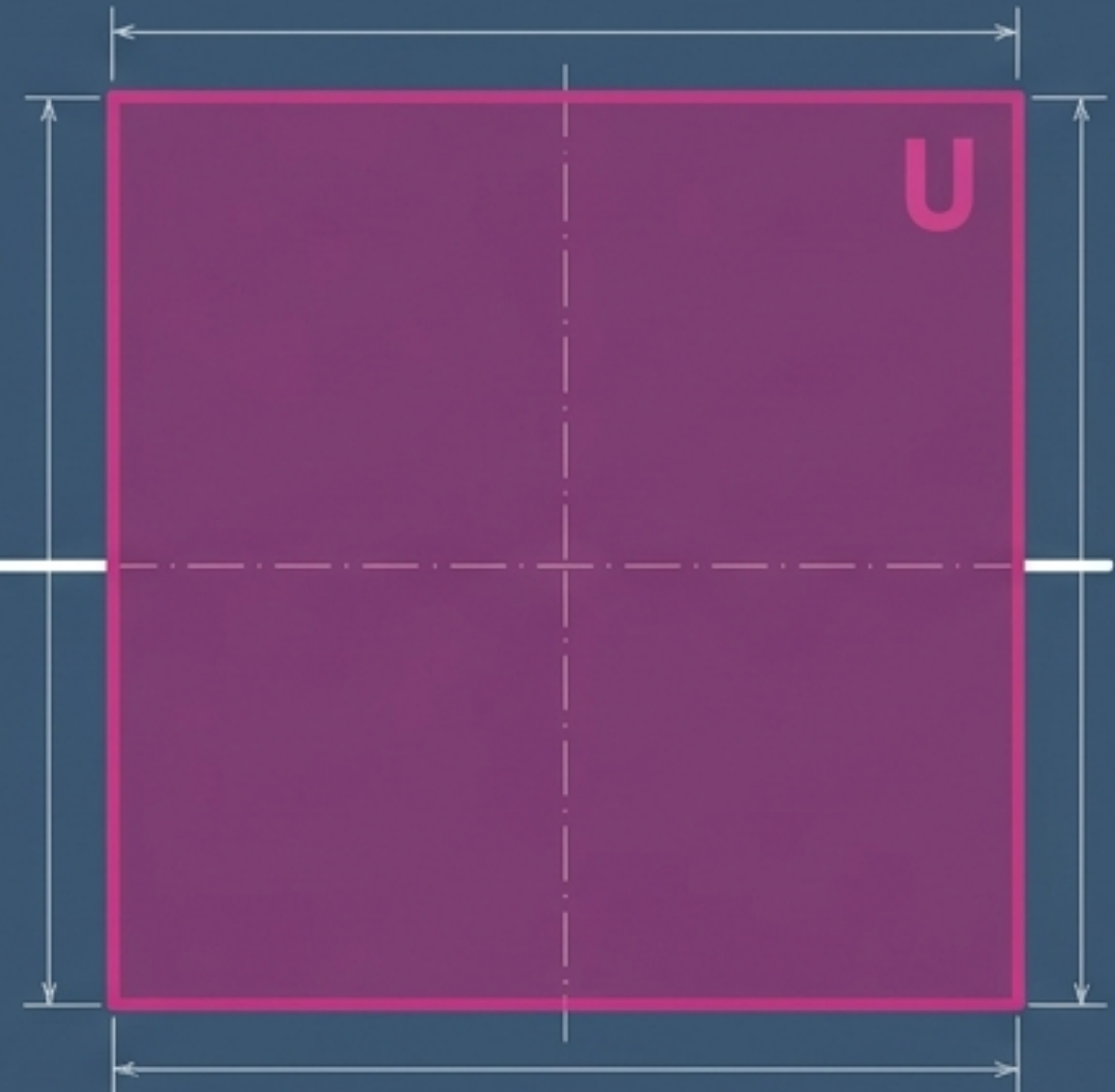


Contains no elements.  
Cardinality:  $|\emptyset| = 0$   
A subset of every set.

## Singleton (Unit) Set

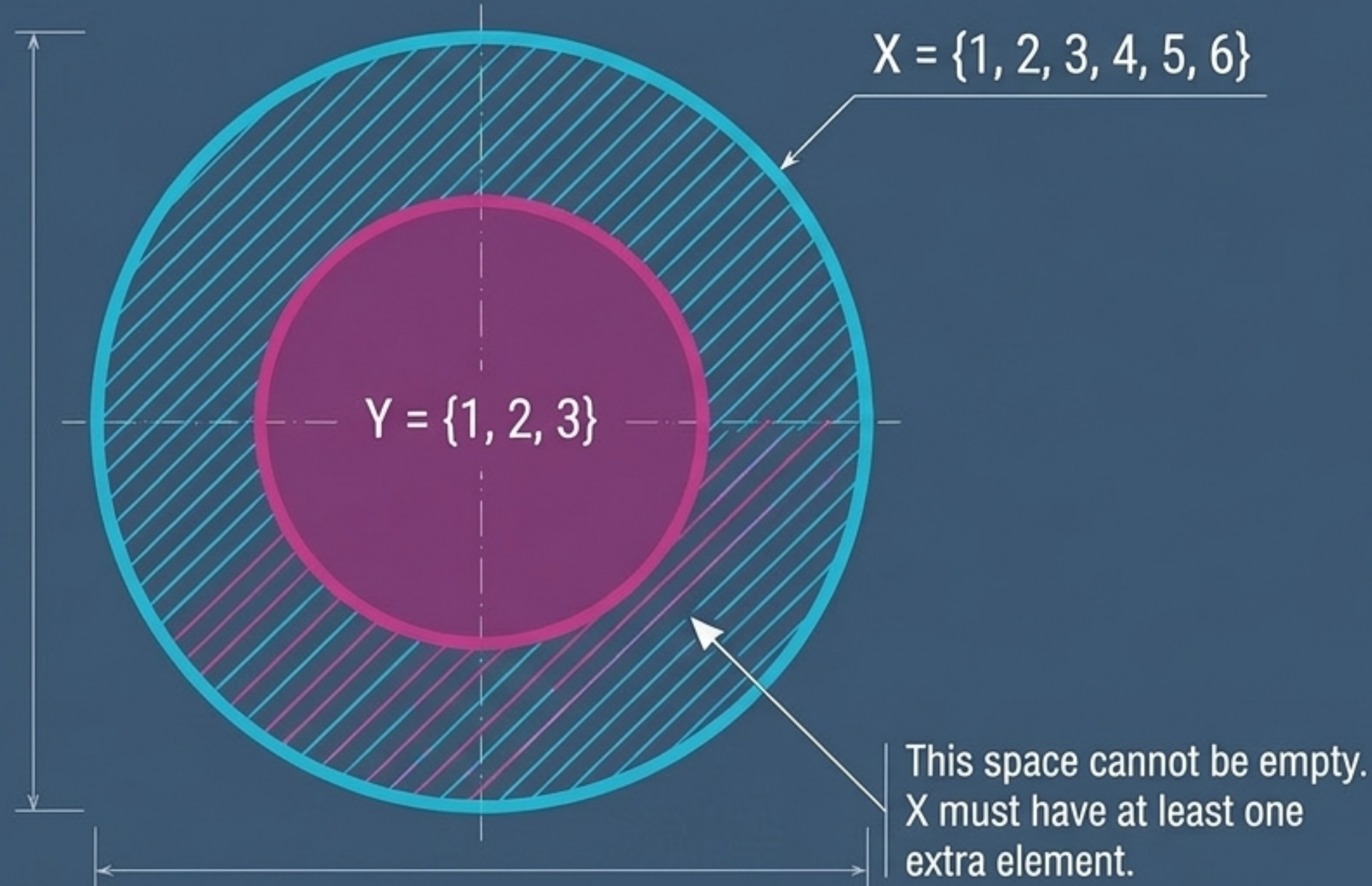


Exactly one element.  
Example:  $S = \{x \in \mathbb{N} \mid 7 < x < 9\} = \{8\}$



The encompassing context.  
The collection of all elements under consideration.  
Every set discussed is a subset of U.

# The Anatomy of Inclusion



## Proper Subset ( $Y \subset X$ )

Every element of Y is also an element of X.

## The Golden Rule

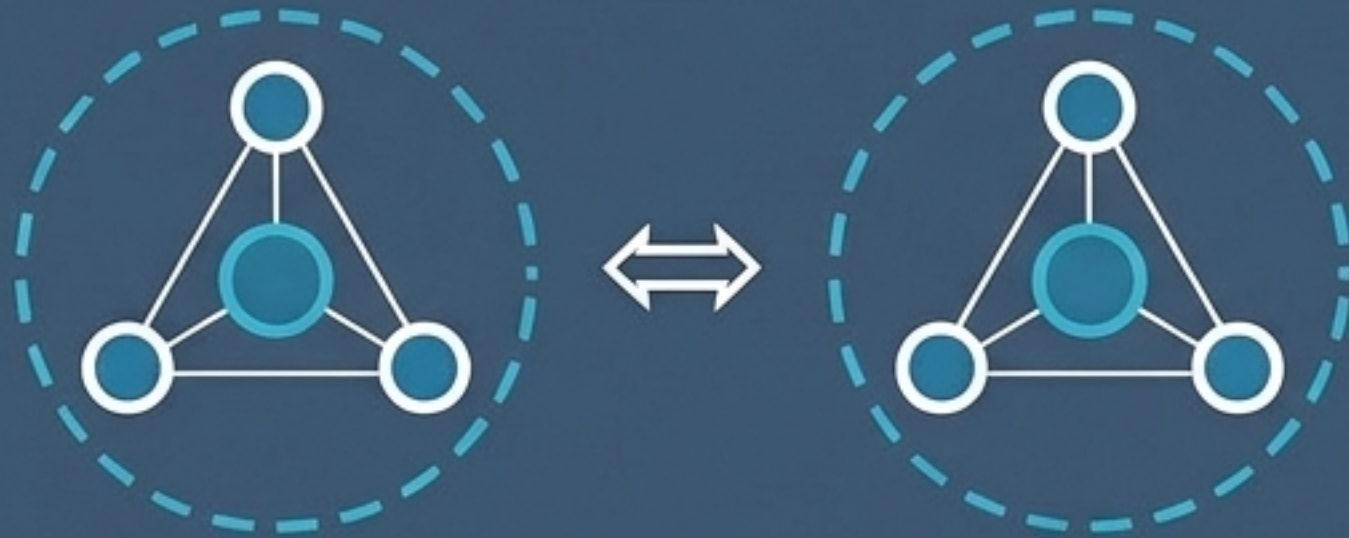
For strict inclusion ( $Y \subset X$ ),  
X and Y cannot be equal.

$$Y \subseteq X \text{ AND } |Y| < |X|.$$

The empty set  $\emptyset$  is a proper subset of every non-empty set.

# Diagnostic Matrix: Equal vs. Equivalent

## Equal Sets

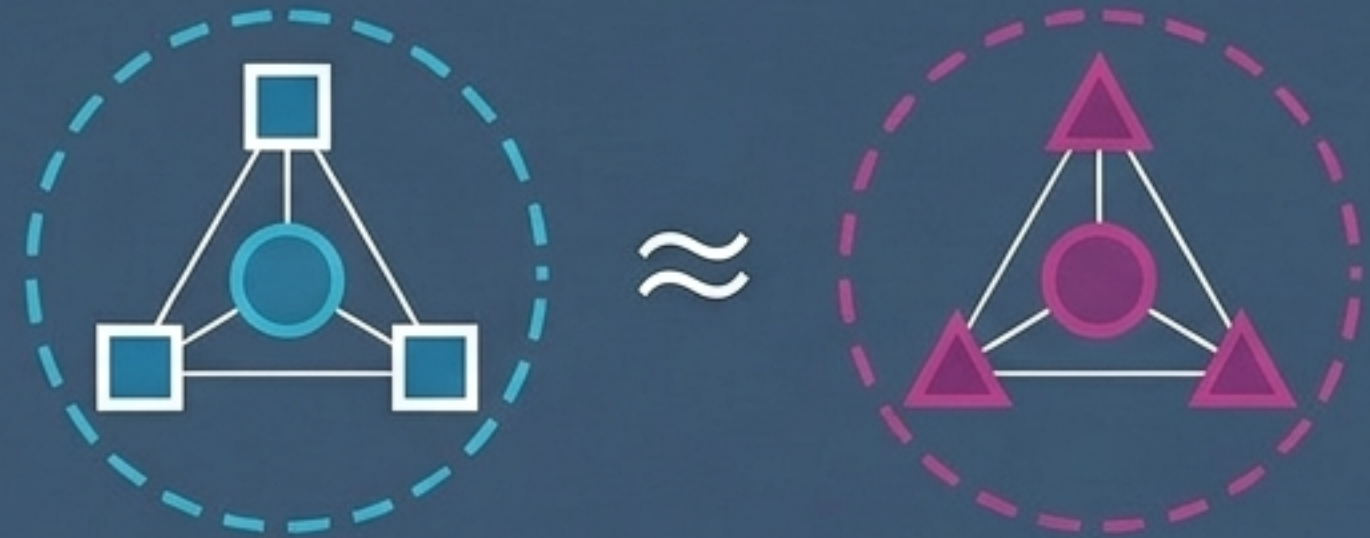


Two sets containing exactly the same elements (order and repetition do not matter).

$$A = B \iff (\forall x)(x \in A \leftrightarrow x \in B)$$

$$\begin{aligned} A &= \{1, 2, 6\} \\ B &= \{6, 1, 2\} \Rightarrow A = B \text{ (and } |A| = |B|) \end{aligned}$$

## Equivalent Sets



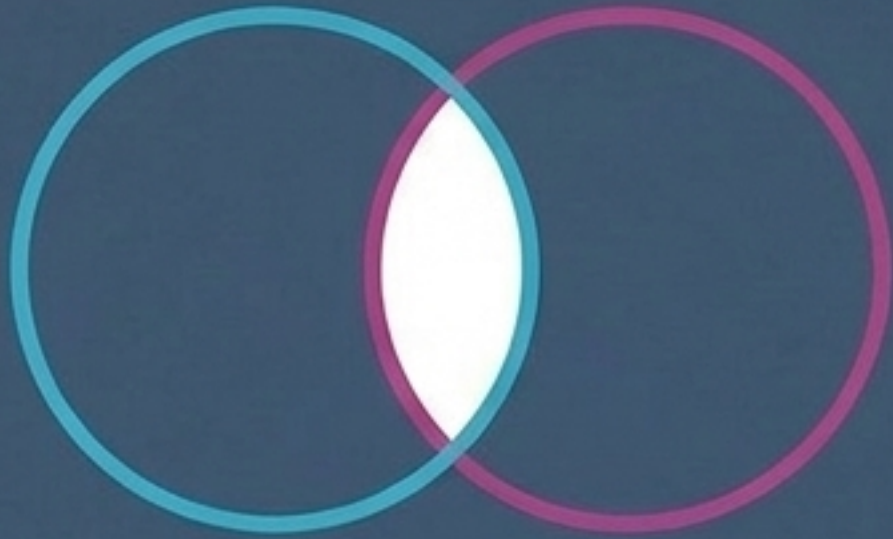
Have the same cardinality ( $|A| = |B|$ ) but may contain different elements.

$$\begin{aligned} A &= \{1, 2, 6\} \\ B &= \{13, 16, 10\} \end{aligned}$$

Both have a cardinality of 3, making them equivalent, but not equal.

# The States of Intersection

## Overlapping Sets

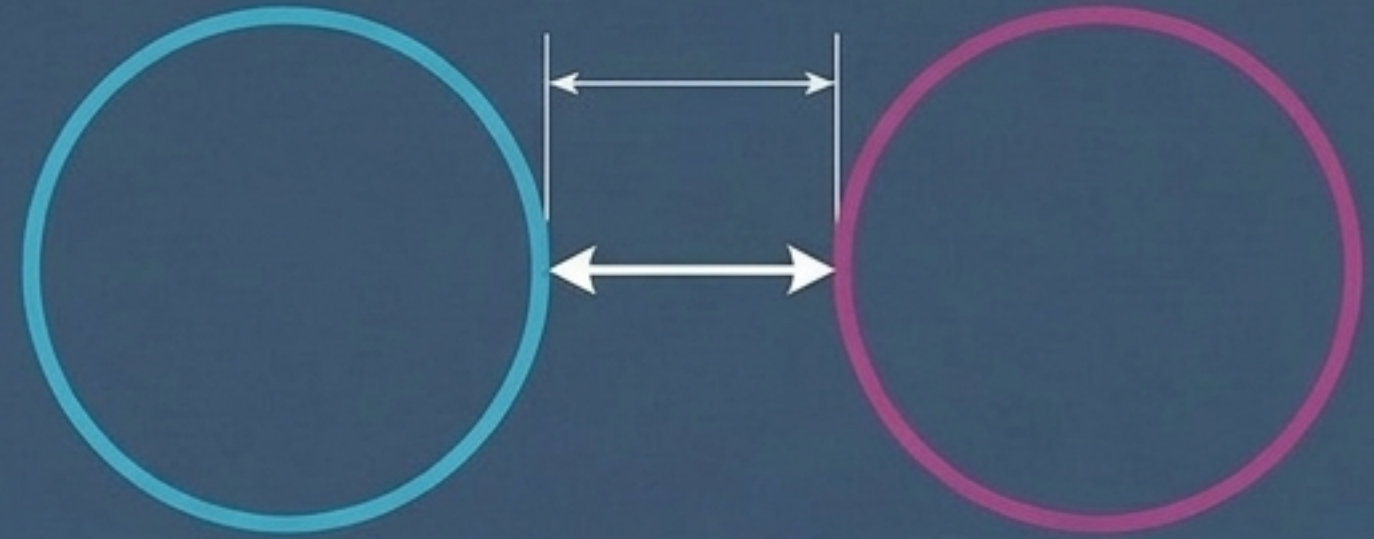


$$A \cap B \neq \emptyset$$

At least one element in common.

$A = \{1, 2, 6\}$   
 $B = \{6, 12, 42\}$   
Common element: 6

## Disjoint Sets



$$A \cap B = \emptyset$$

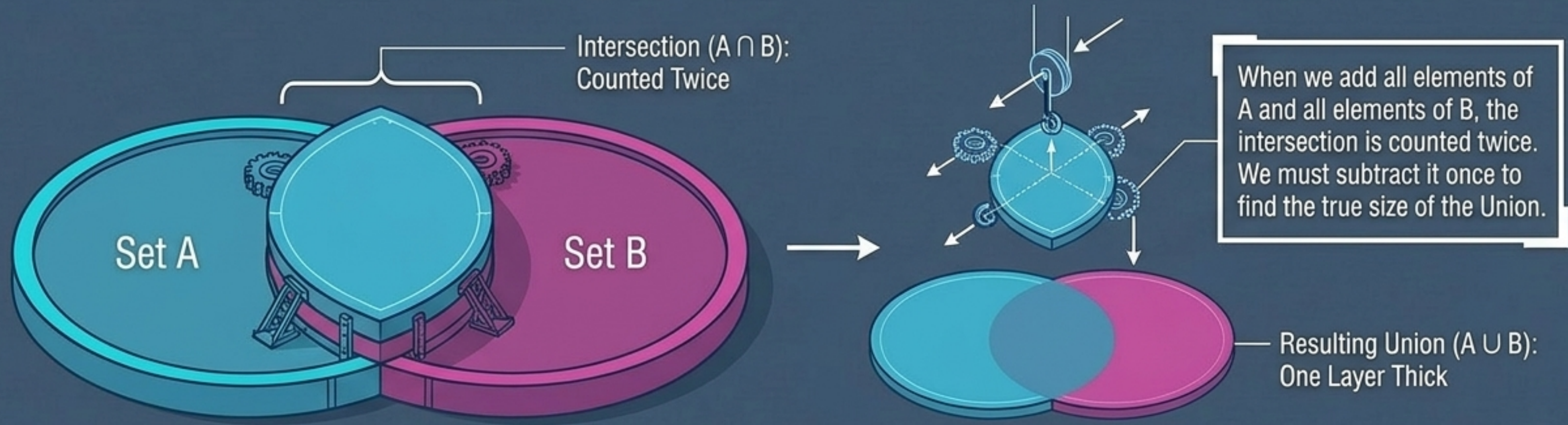
No elements in common.

Simplified Cardinality:  
 $n(A \cup B) = n(A) + n(B)$

$A = \{1, 2, 6\}, \quad B = \{7, 9, 14\}$

# Visualizing the Inclusion-Exclusion Principle

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

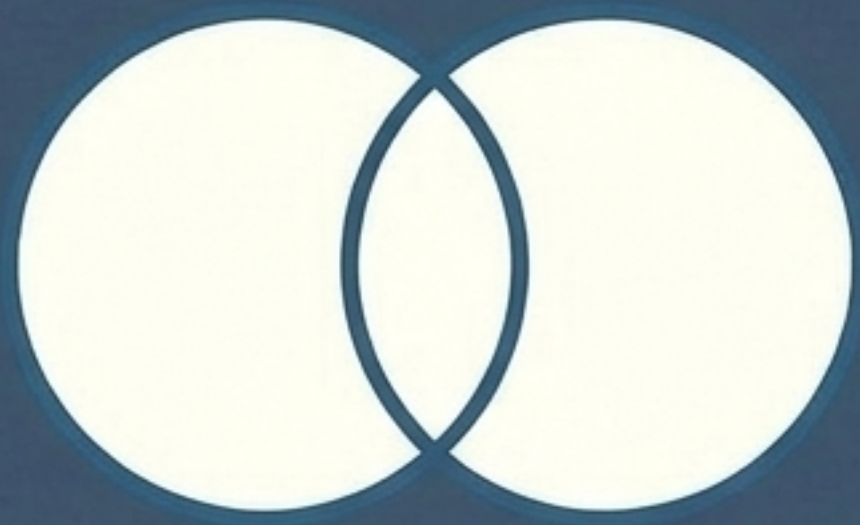


Alternative Formulations:

$$n(A \cup B) = n(A - B) + n(B - A) + n(A \cap B)$$
$$n(A) = n(A - B) + n(A \cap B)$$

# Core Operations I: Synthesis

## Union ( $A \cup B$ )

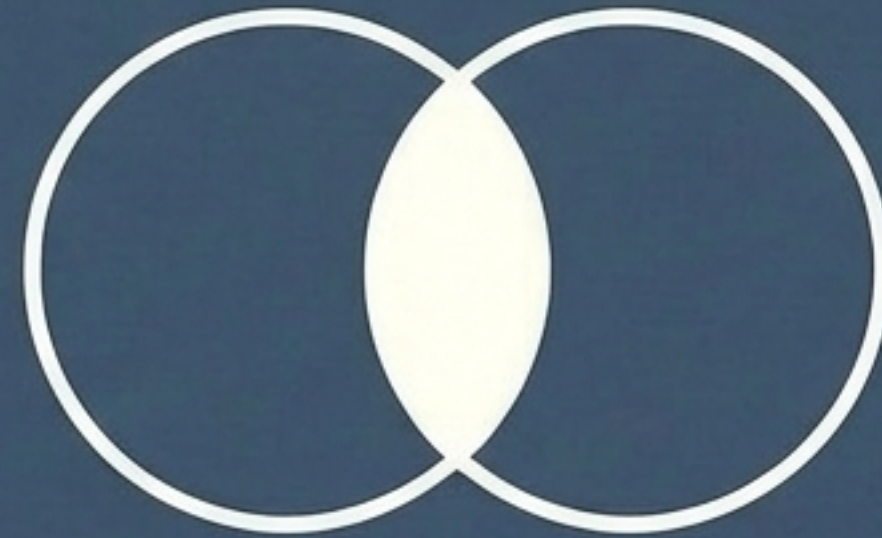


The set of all elements that belong to A, to B, or to both. Logical OR ( $\vee$ ).

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

$$A = \{10, 11, 12, 13\}, B = \{13, 14, 15\} \\ \Rightarrow A \cup B = \{10, 11, 12, 13, 14, 15\}$$

## Intersection ( $A \cap B$ )



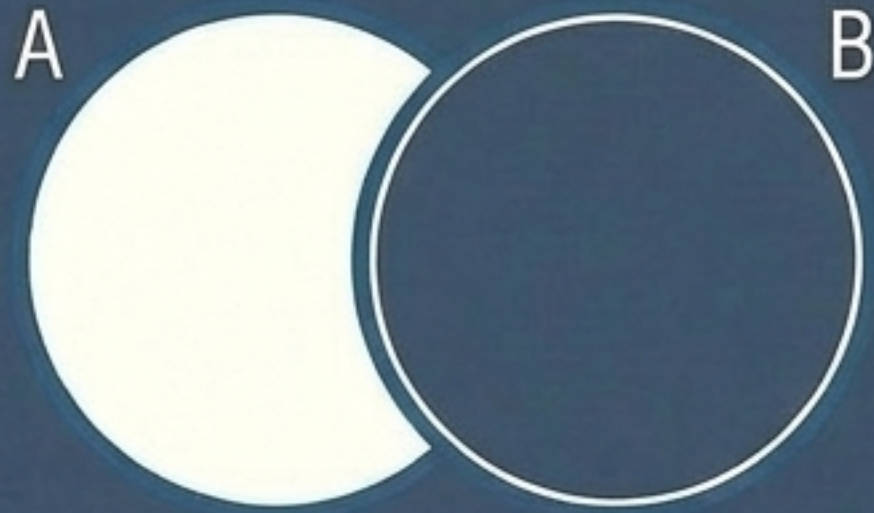
Elements that belong to both A and B. Logical AND ( $\wedge$ ).

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

$$A = \{11, 12, 13\}, B = \{13, 14, 15\} \\ \Rightarrow A \cap B = \{13\}$$

# Core Operations II: Removal

## Set Difference ( $A - B$ )



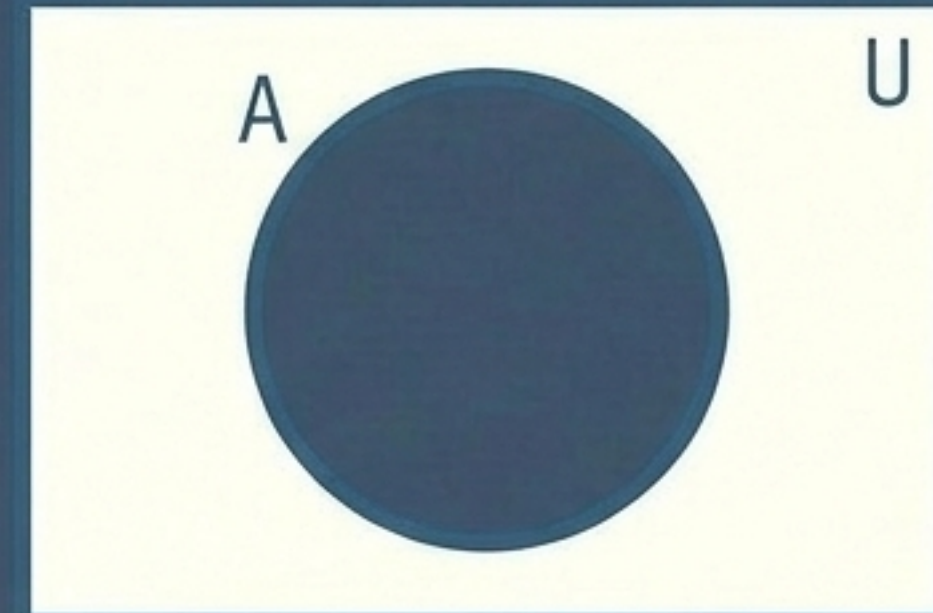
Elements in A but not in B.

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$

Critical Note:  $A - B \neq B - A$  in general.

If  $A = \{10, 11, 12, 13\}$  and  $B = \{13, 14, 15\}$ , then  $A - B = \{10, 11, 12\}$ , but  $B - A = \{14, 15\}$ .

## Complement ( $A'$ or $A^c$ )

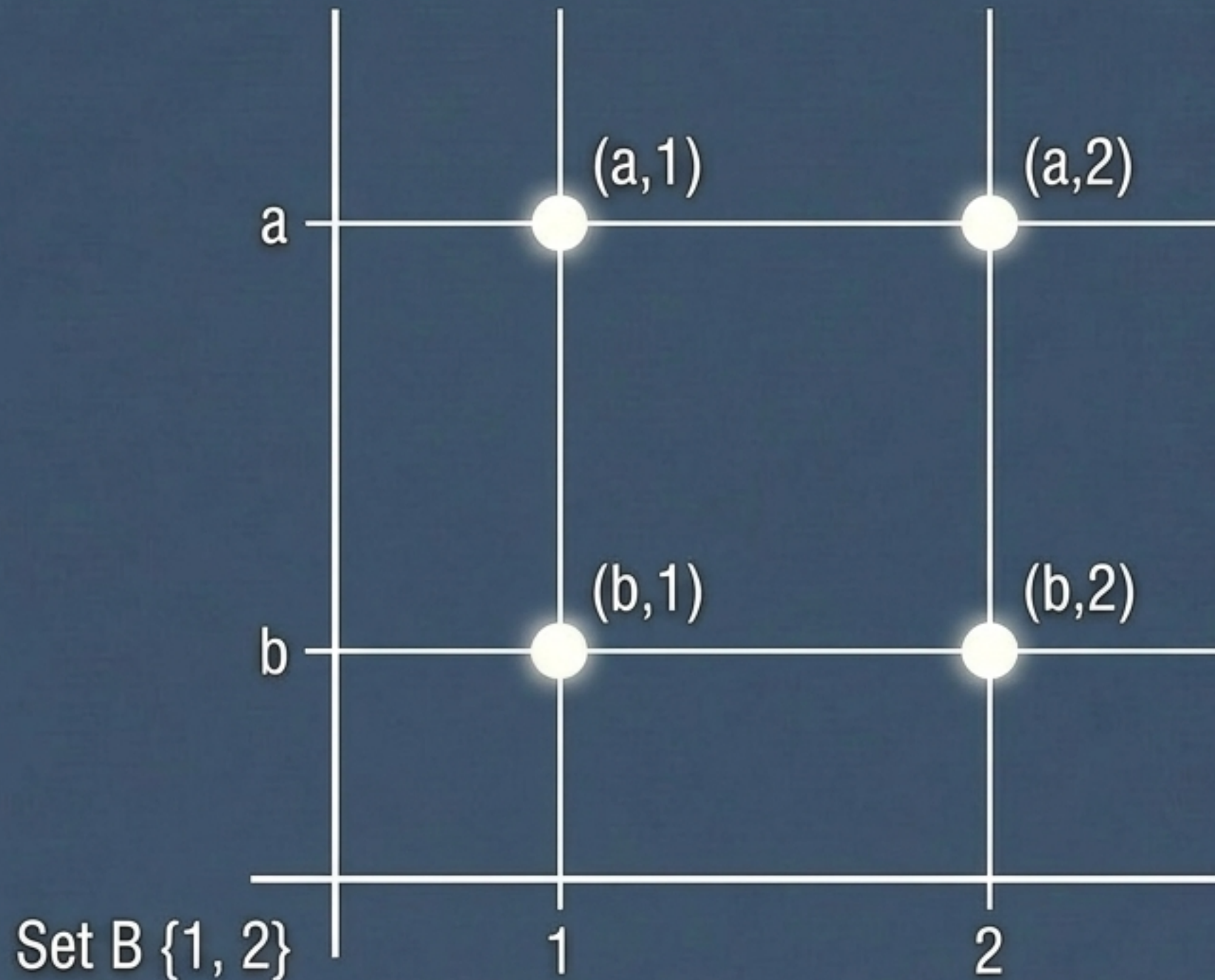


Elements of U not in A.

$$A' = \{x \in U \mid x \notin A\}$$

If  $U = \{10, 11, 12, 13, 14, 15\}$  and  $A = \{11, 12, 13\}$ , then  $A' = \{10, 14, 15\}$ .

# The Multiplier: Cartesian Products



## Definition

The Cartesian Product ( $A \times B$ ) is the set of all ordered pairs  $(x, y)$  where  $x \in A$  and  $y \in B$ .

## The Grid Result

$$A \times B = \{(a,1), (a,2), (b,1), (b,2)\}$$

## Crucial Warning

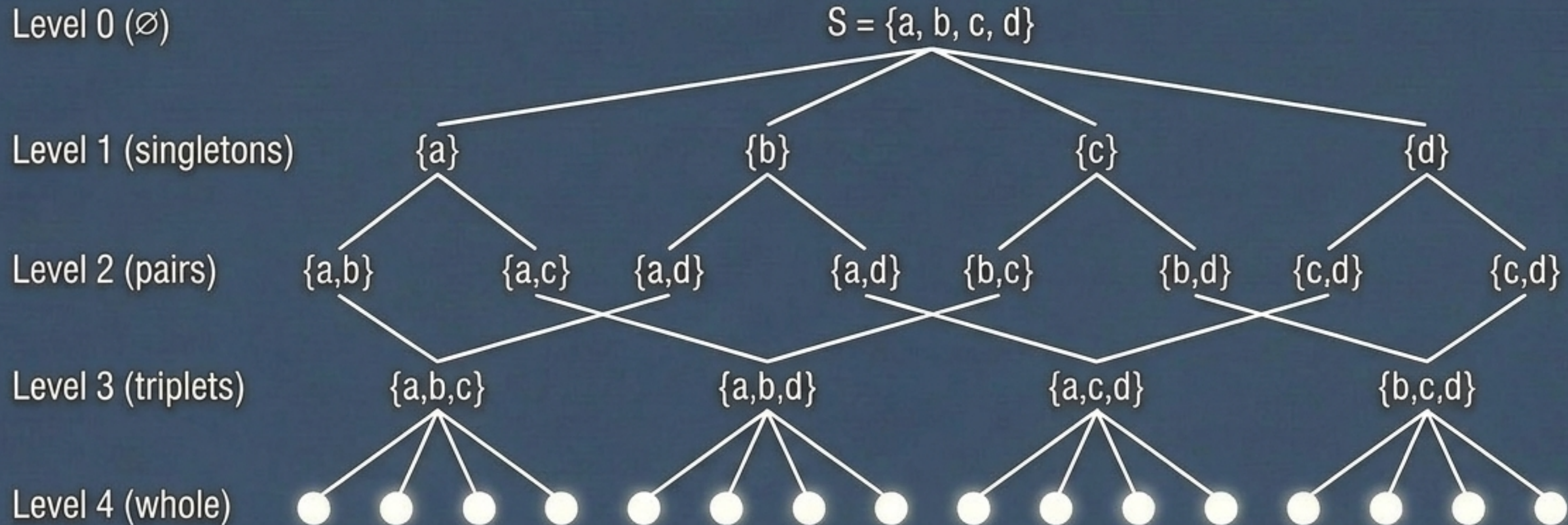
Order matters intrinsically.

$B \times A = \{(1,a), (1,b), (2,a), (2,b)\}$ , which is a completely different set of coordinates.

Note: We often write  $A^2$  for  $A \times A$ .

For  $n$  sets ( $A_1 \times A_2 \times \dots \times A_n$ ), this creates  $n$ -tuples.

# The Power Set: A Universe of Subsets



## Definition:

The power set of  $S$ , denoted  $P(S)$  or  $2^S$ , is the set of all subsets of  $S$ , including  $\emptyset$  and  $S$  itself.

## Rule:

The Exponential Rule:  
If  $|S| = n$ , then  $|P(S)| = 2^n$

## Calculation:

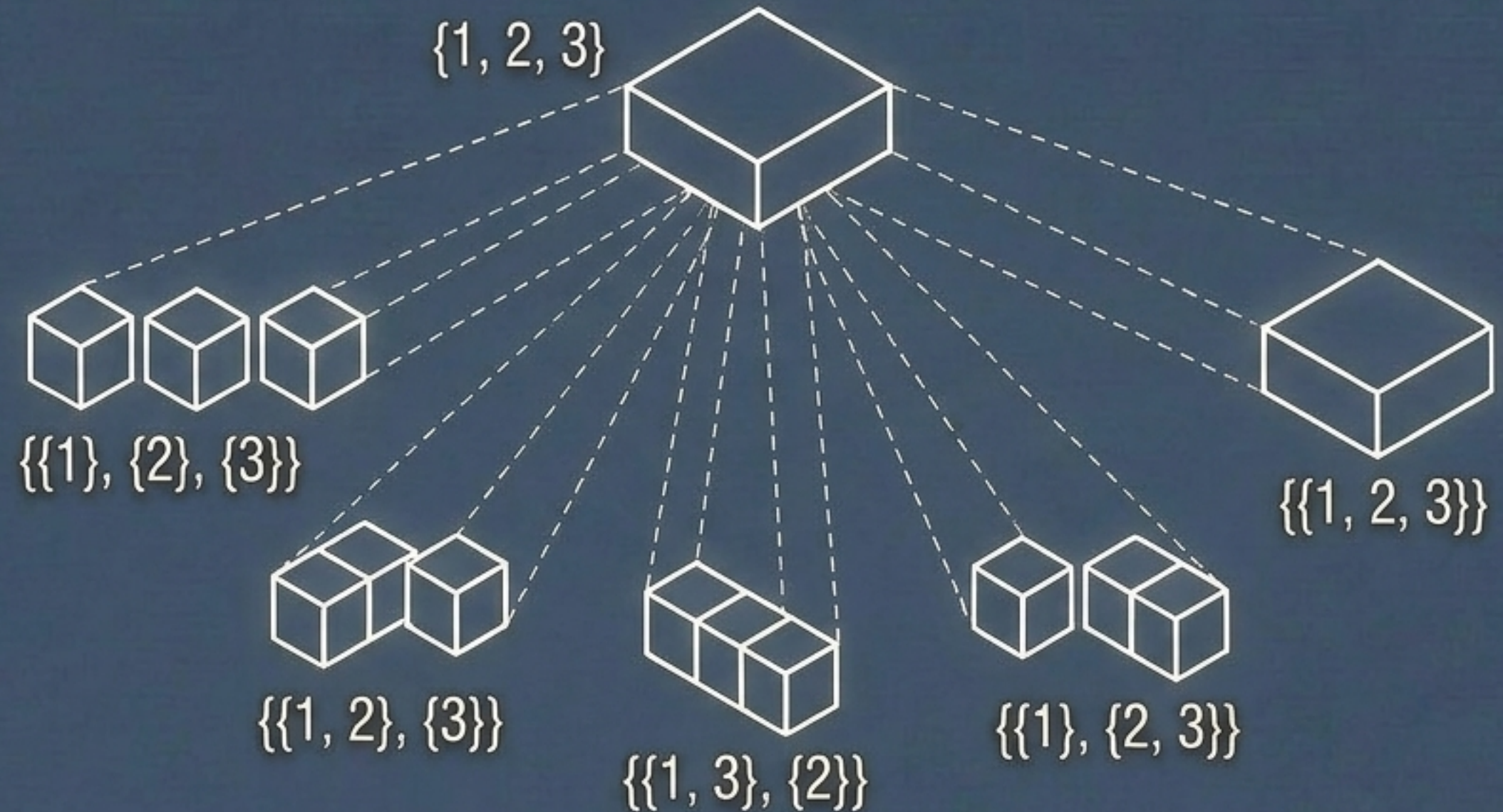
The Mathematical Explosion:  
 $|S| = 4 \Rightarrow |P(S)| = 2^4 = 16$ . This simple 4-element set generates 16 entirely distinct subsets.

# Breaking It Down: Partitions & Bell Numbers

## The 3 Rules of Partitions

A partition  $P$  of set  $S$  must be:

1. Non-empty ( $P_i \neq \emptyset$ )
2. Pairwise disjoint ( $P_i \cap P_j = \emptyset$ )
3. Union equals the whole set ( $P_1 \cup P_2 \cup \dots = S$ )



**Bell numbers ( $B_n$ ) scale rapidly:**  
 $B_0=1, B_1=1, B_2=2, B_3=5, B_4=15, B_5=52$

# Bridging the Gap: Binary Relations

## The Synthesis

A **binary relation**  $R$  from set  $A$  to set  $B$  is not a new abstract concept—it is simply a subset of the Cartesian product  $A \times B$ .

## Notation

We write  $a R b$  (or  $(a, b) \in R$ ) when  $a$  is related to  $b$ .

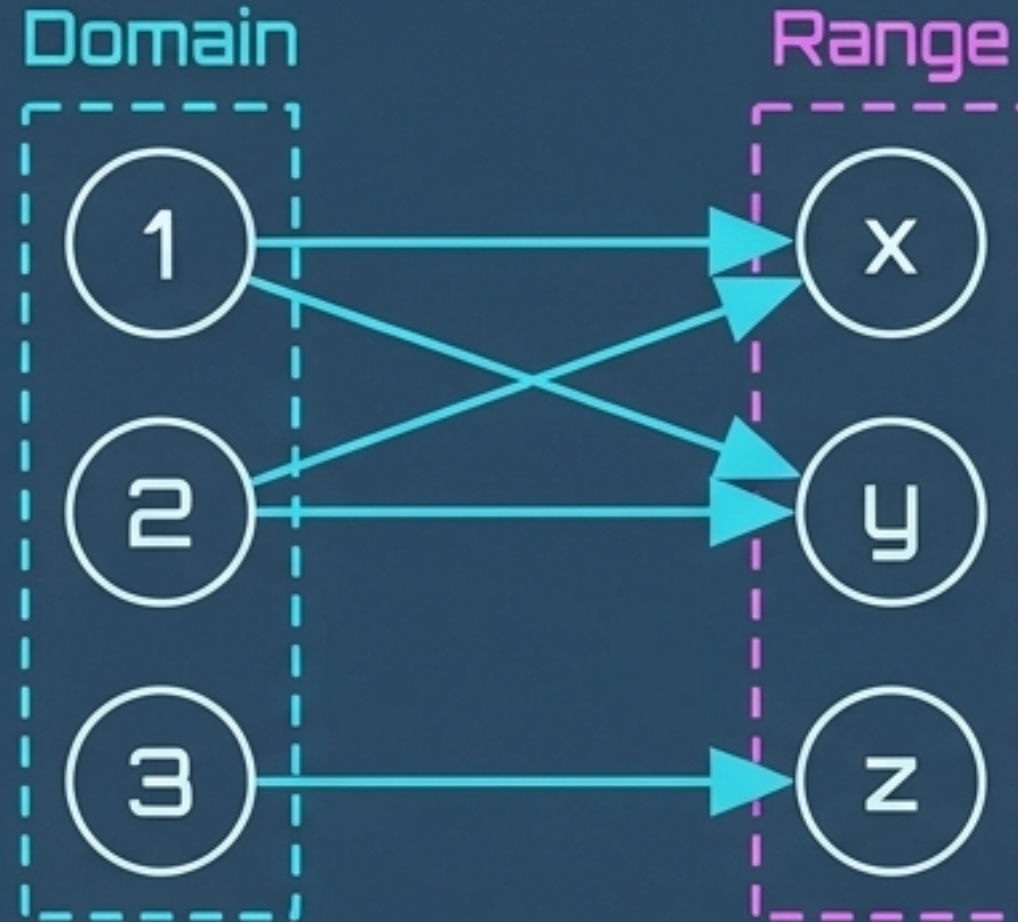
$$R \subseteq A \times B$$

## Self-Relation

If  $R \subseteq A \times A$ ,  $R$  is a relation on  $A$ .

Both the empty relation and the universal relation ( $A \times A$ ) are always possible.

# Anatomy of a Relation: Domain & Range



## Domain of $R$

The first components that actually appear.

$$\{a \in A \mid \exists b \in B \text{ such that } (a, b) \in R\}$$

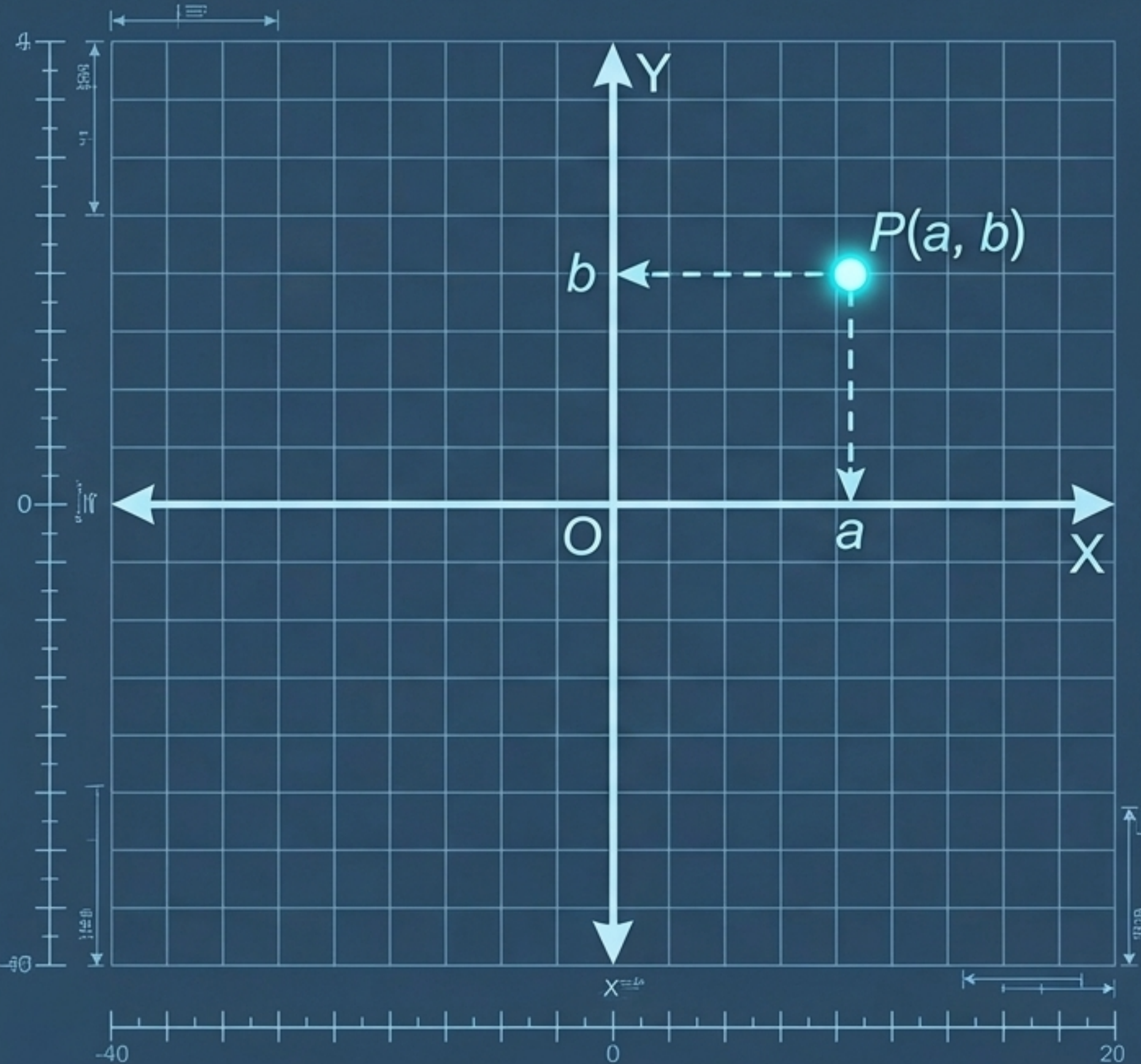
## Range of $R$

The second components that appear.

$$\{b \in B \mid \exists a \in A \text{ such that } (a, b) \in R\}$$

**Set Inclusion Note:** For any collection of sets, the relation  $\subseteq$  (Set Inclusion) is a binary relation. It is a partial order: reflexive, antisymmetric, and transitive.

# The Geometric View: The Cartesian Plane ( $\mathbb{R}^2$ )



Grounding abstract sets in physical geometry.  $\mathbb{R}$  denotes the set of real numbers.

**The Ultimate Cross Product:**

$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$  is the set of all ordered pairs of real numbers.

**Mapping:**

Each point  $P$  in the plane represents a specific ordered pair  $(a, b)$ . The vertical line through  $P$  meets the  $x$ -axis at  $a$ ; the horizontal line meets the  $y$ -axis at  $b$ .

# The Master Blueprint (Quick Revision)



## Subsets & Scale

$X \subset Y \Rightarrow X \subseteq Y$  and  $X \neq Y$ .  
Singleton = 1 element.  
 $|\emptyset| = 0$ .



## Identities

Equal Sets = identical elements.  
Equivalent Sets = identical cardinality.



## Intersection Logic

Overlapping  $\Rightarrow A \cap B \neq \emptyset$ .  
Disjoint  $\Rightarrow A \cap B = \emptyset$ .



## Cardinality

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$



## Operations

Complement  $A' = U - A$ .  
Difference  $A - B = \{x \in A \mid x \notin B\}$ .  
(Remember  $A - B \neq B - A$ ).



## Permutations

Power Set  $|P(S)| = 2^n$ .  
Bell Number  $B_n$  = partitions of  $n$ -element set ( $B_3 = 5$ ).



## Relations

$R$  from  $A$  to  $B$  is a subset of  $A \times B$ .  
**Domain** = first components.  
**Range** = second components.  
**Order matters** in  $A \times B$ .